

1. Setup

$$x_0 \in \mathbb{C}^{H \times W}$$

complex lateral field; H depth, W A-lines

$$\mathbf{x}_0[0] = \operatorname{Re} x_0, \quad \mathbf{x}_0[1] = \operatorname{Im} x_0$$

two-channel real packing for the network

$$(\mathcal{F}x)[h, k] = \frac{1}{\sqrt{W}} \sum_{w=0}^{W-1} x[h, w] e^{-2\pi i k w / W}$$

unitary DFT, lateral axis only

$$\hat{f}_j = (j - c)/W \in [-\tfrac{1}{2}, \tfrac{1}{2}), \quad c = \lfloor W/2 \rfloor$$

normalized frequency after fftshift S

2. Sampling theory

$$Y[\kappa] = \frac{1}{K} \sum_{p=0}^{K-1} X[\kappa + pN'], \quad N' = W/K$$

decimation by K periodizes the spectrum

$$\text{HWHM}_K \approx \min(K \cdot \text{HWHM}_0, 0.5)$$

normalized bandwidth stretches, then saturates

$$K^* = 0.5 / \text{HWHM}_0$$

critical factor: aliasing begins here

3. Three degradations

$$\text{A: } \tilde{x} \mapsto G_\sigma \odot \tilde{x}$$

Gaussian mask = resolution loss (defocus, NA)

$$\text{B: } \tilde{x} \mapsto M_\rho \odot \tilde{x}$$

brick-wall = ideal anti-aliased downsampling

$$\text{C: } \tilde{x} \mapsto M_{1/K} \odot \frac{1}{K} \sum_{p=0}^{K-1} \mathcal{T}_{pW/K} \tilde{x}$$

fold-in + brick-wall = true sub-Nyquist sampling

4. Forward operator

$$\tilde{y}_t = \tilde{x}_0 + \alpha(t) \sum_{p=1}^{K-1} \mathcal{T}_{pW/K} \tilde{x}_0$$

fold-in: shift by pW/K bins = shift by p/K in $\hat{f}$

$$D(x_0, t) = g(t) \mathcal{F}^{-1} S^{-1} [M_{\rho(t)} \odot \tilde{y}_t]$$

then window to the retained band

$$\rho(t) = 1 - (1 - 1/K) t/T, \quad \alpha(t) = t/T, \quad g(t) = [1 + \alpha(t)(K - 1)]^{-1}$$

$$D(x, 0) = x \quad D(x, T) = \text{measurement}$$

verified to 10^{-16} ; requires $K \mid W$

5. Training

$$t \sim \mathcal{U}\{1, \dots, T\}, \quad \mathbf{x}_t = D(\mathbf{x}_0, t)$$

$$\mathcal{L}(\theta) = \mathbb{E}_{\mathbf{x}_0, t} \| R_\theta(D(\mathbf{x}_0, t), t) - \mathbf{x}_0 \|_1$$

network predicts x_0 directly

$$+ \lambda \mathbb{E} [w \cdot (1 - \cos(\hat{\theta} - \theta))], \quad w = |x_0|$$

optional amplitude-weighted phase term

6. Reverse process

$$\hat{\mathbf{x}}_0 = R_\theta(\mathbf{x}_t, t)$$

$$\mathbf{x}_{t-1} = \mathbf{x}_t - D(\hat{\mathbf{x}}_0, t) + D(\hat{\mathbf{x}}_0, t-1)$$

systematic error of R_θ cancels between steps

$$\text{DC}(\hat{x}, y) = \mathcal{F}^{-1} S^{-1} [(1 - M) \odot S\mathcal{F}\hat{x} + M \odot S\mathcal{F}y]$$

inference only -- never during training